USN. | | | | | | | | | | [24MTAD102]

FIRST SEMESTER M.Tech EXAMINATION, JANUARY- 2025

FLIGHT MECHANICS I

Time: 3 Hours **Maximum Marks: 100**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
ii. Missing data may be suitably assumed.

			CO's	Blooms Level	Marks
1.	**a.**	Explain the following. **i.** Calibrated air speed **ii.** Equivalent air speed **iii.** Power specific fuel consumption rate **iv.** Power variation with altitude **v.** Induced drag	**CO1**	**2**	2+(4 *3) =14
	b.	Classify different types of drag encountered by an aircraft. Explain how to obtain drag polar for an aircraft.	**CO1**	**2**	**06**
2.	**a.**	An aircraft weighing 24000 N has wing area of 80 m2 and max speed of 120 m/s at sea level. The maximum lift coefficient with flaps down is 2.2 **i.** If C_{Lmax} flaps up=1.5, below what TAS, must flaps be used at sea level and at 6100 m, where σ=0.533 **ii.** Plot a curve showing variation of C_L for straight and level flight over the feasible speed range at sea level	**CO2**	**3**	**10**
	b.	Derive the expression for minimum drag and minimum power condition if the drag polar is given by $C_D = C_{D0} + k\,C_L^2$ What is the value of C_D and E, aerodynamic efficiency?	**CO2**	**2**	**07** **03**
3.	**a.**	Obtain and plot the variation of drag with flight speed for straight and level flight. What happens to the drag curve with change in altitude? Plot the variation of drag with VEAS. What is the relation between V_{TAS} and V_{EAS}?	**CO2**	**2**	**10**
	b.	A sail plane weighs 4500 N and has wing loading of 600 N/m^2. The drag polar is given by C_d= 0.01+ 0.022C_L^2. The sail plane is launched from 350 m altitude in still air. Find the following assuming sea level	**CO3**	**3**	**10**

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		density: **i.** Greatest horizontal distance covered by the sail plane, **ii.** greatest duration of flight **iii.** Find out the corresponding flight speed.			
		OR			
4.	**a.**	Derive the equations of motion of climbing flight and obtain the expression for climb angle and rate of climb. From this, derive the expression for steepest climb.	**CO4**	**2**	**10**
	b.	Based on small angle approximation for gliding flight, show that the greatest flight duration occurs at minimum power. Obtain the corresponding flight speed and aerodynamic efficiency.	**CO3**	**2**	**10**
5.	**a.**	Obtain the expression for range for a jet powered aircraft based on constant altitude and constant lift coefficient. What happens for a head or tail wind on range?	**CO5**	**2**	**10**
	b.	An aircraft weighing 240000 N has wing area of 80 m^2 and maximum speed of 120 m/s at sea level. Maximum lift coefficient with flaps down is 2.2. i. If C_{Lmax} (flaps up) = 1.5, below what true air speed, flaps must be used at sea level and at 6100 m, where the density ratio σ =0.533 ii. Plot a curve showing variation of C_L for straight and level flight over the feasible speed range at sea level. iii) Obtain the minimum drag for the aircraft if $(L/D)_{max}$ = 18.	**CO4**	**3**	**10**
		OR			
6.	**a.**	Explain the following- i. Breguet range formula ii. Safe range iii. Endurance expression for turbojet aircraft iv. Absolute Ceiling v. Endurance	**CO5**	**2**	**(4x2.5)+2=12**
	b.	Obtain the expression for endurance for an aircraft with jet engines.	**CO5**	**2**	**08**
7.	**a.**	Discuss and explain the various critical speeds during take-off. What is the significance of screen height during take-off? Illustrate take-off performance using speed versus take-off distance diagram.	**CO6**	**2**	**10**

	b.	What is the difference between takeoff run and take off distance? Obtain the equation for take-off distance.	**CO6**	**2**	**10**
		OR			
8.	**a.**	Define load factor, turn rate and radius of turn and obtain the expressions for load factor, turning rate and turning radius.	**CO6**	**2**	**10**
	b.	Define or explain the following- **i.** Engine failure speed V_1 during take-off **ii.** Tightest turn **iii.** turn rate **iv.** Load factor **v.** take-off run	**CO6**	**2**	**10**